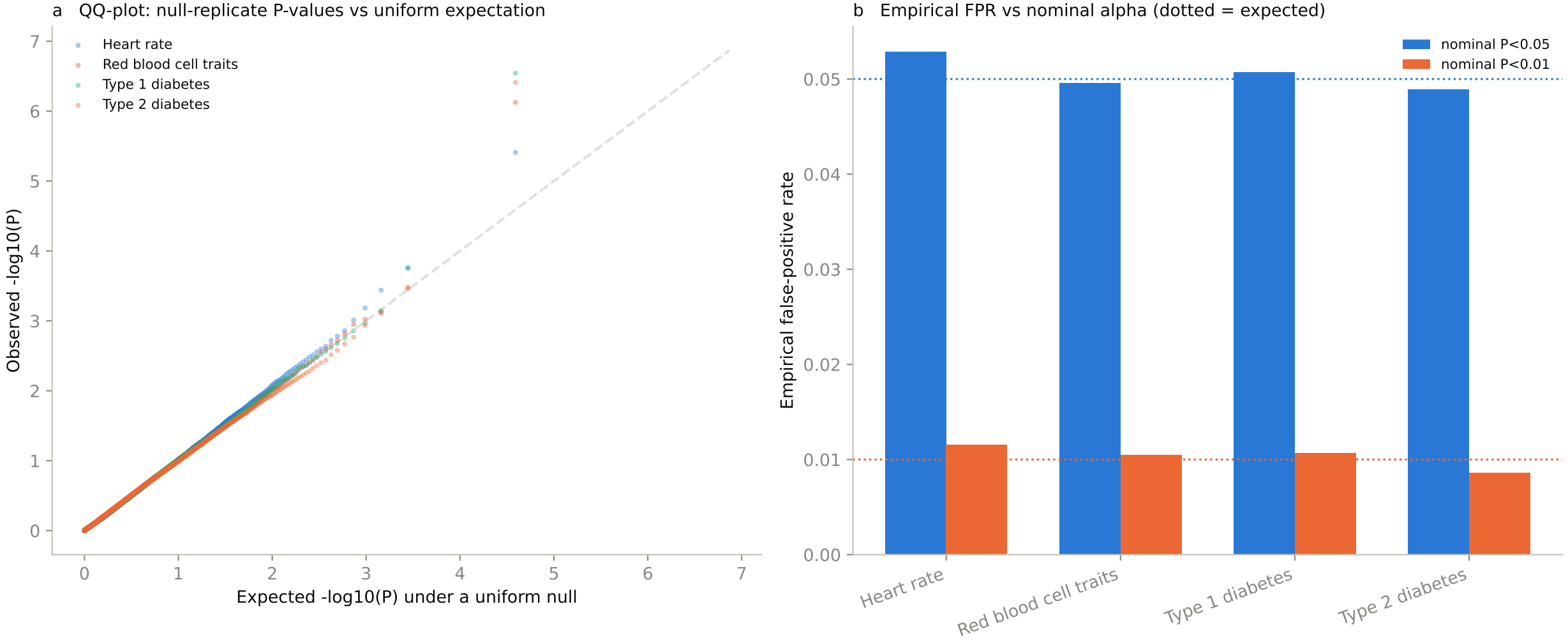

a QQ-plot: null-replicate P-values vs uniform expectation
Heart rate
Red blood cell traits
Type 1 diabetes
Type 2 diabetes
Observed -log10(P)
Expected -log10(P) under a uniform null
0
1
2
3
4
5
6
7
b Empirical FPR vs nominal alpha (dotted = expected)
nominal P<0.05
nominal P<0.01
Empirical false-positive rate
0.00
0.01
0.02
0.03
0.04
0.05
Heart rate
Red blood cell traits
Type 1 diabetes
Type 2 diabetes